



# Incident handler's journal

## Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

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| <b>Date:</b><br>May 28, 2026 | <b>Entry:</b><br>1                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Description</b>           | Documenting a ransomware attack on a small healthcare clinic that disrupted access to patient records and critical software systems.                                                                                                                                                                                                                                                                                                                                                      |
| <b>Tool(s) used</b>          | <ul style="list-style-type: none"><li>• SIEM (Security Information and Event Management) for log analysis</li><li>• Antivirus and endpoint detection tools</li><li>• Email filtering system</li><li>• Incident response playbook</li></ul>                                                                                                                                                                                                                                                |
| <b>The 5 W's</b>             | <p><b>Who caused the incident:</b><br/>An organized cybercriminal group known for targeting healthcare and transportation sectors.</p> <p><b>What happened:</b><br/>Employees received phishing emails containing malicious attachments. Opening the attachment installed malware that encrypted files and displayed a ransom note demanding payment for decryption.</p> <p><b>When did the incident occur:</b><br/>Tuesday at 9:00 a.m.</p> <p><b>Where did the incident happen:</b></p> |

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|                  | <p>At the clinic's main office network, affecting all connected workstations and servers.</p> <p><b>Why did the incident happen:</b></p> <p>Attackers exploited human error through phishing to gain access and deploy ransomware, aiming for financial gain.</p>                                                                                                                                                                                                    |
| Additional notes | <ul style="list-style-type: none"> <li>• The clinic immediately shut down affected systems and contacted external cybersecurity specialists.</li> <li>• This incident highlights the importance of employee phishing awareness training, regular backups, and network segmentation.</li> <li>• Future prevention steps include implementing multi-factor authentication, updating email filters, and conducting tabletop exercises for incident response.</li> </ul> |

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| <b>Date:</b><br>May 28, 2026 | <b>Entry:</b><br>2                                                                                                                                                                                                                                                                                                                                                            |
| Description                  | This lab focused on analyzing network traffic using Wireshark and tcpdump. I captured and inspected ICMP, DNS, and TCP packets to identify normal and potentially suspicious activity. The exercise demonstrated how packet captures reveal reconnaissance behavior, domain resolution, and HTTP communication patterns that analysts use to detect and respond to incidents. |
| Tool(s) used                 | Wireshark, tcpdump                                                                                                                                                                                                                                                                                                                                                            |
| The 5 W's                    | <p><b>Who caused the incident?</b></p> <p>A simulated external attacker probing the network from IP 142.250.1.139 and normal internal network operations generating DNS and HTTP traffic.</p>                                                                                                                                                                                 |

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|                  | <p><b>What happened?</b></p> <p>ICMP echo requests indicated potential reconnaissance; DNS queries resolved <i>opensource.google.com</i>; TCP packets showed HTTP communication to IP 142.250.1.139 (port 80).</p> <p><b>When did the incident occur?</b></p> <p>May 28, 2026, during the “Analyze your first packet” lab session.</p> <p><b>Where did the incident happen?</b></p> <p>Within the controlled virtual machine environment used for packet capture and analysis.</p> <p><b>Why did the incident happen?</b></p> <p>The lab simulated real-world network activity to teach how analysts detect reconnaissance, data exfiltration, and normal traffic patterns using packet analysis tools.</p> |
| Additional notes | <p>This exercise reinforced how packet captures and protocol analyzers help identify abnormal network behavior early in the attack chain. Future prevention steps include implementing ICMP rate limiting, monitoring DNS queries for anomalies, and correlating TCP traffic with SIEM alerts for deeper threat detection.</p>                                                                                                                                                                                                                                                                                                                                                                              |

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| <b>Date:</b><br>May 28, 2026 | <b>Entry:</b><br>3                                                                                                                                                                                                                                                                                                              |
| Description                  | <p>In this lab, I used tcpdump on a Linux virtual machine to identify network interfaces, capture live network traffic, save it to a packet capture file, and analyze the captured data. The activity demonstrated how security analysts use tcpdump to inspect packet headers, filter traffic, and examine hexadecimal and</p> |

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|                  | ASCII packet contents for deeper forensic analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Tool(s) used     | tcpdump, curl, Linux terminal utilities (ifconfig, ls)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| The 5 W's        | <p><b>Who caused the activity?</b></p> <p>Normal system processes and manually generated HTTP traffic from the Linux VM. No malicious actor was involved; all traffic was intentionally generated for training.</p> <p><b>What happened?</b></p> <p>I identified available network interfaces, captured live traffic from eth0, filtered packets, saved HTTP traffic to capture. pcap, and analyzed packet headers and payloads using verbose and hex/ASCII output modes.</p> <p><b>When did it occur?</b></p> <p>During the tcpdump lab session on May 28, 2026.</p> <p><b>Where did it occur?</b></p> <p>Inside the Linux virtual machine provided by the lab environment.</p> <p><b>Why did it happen?</b></p> <p>To practice real-world packet capture and analysis techniques used in detection and response, including identifying interfaces, filtering traffic, and interpreting packet structure.</p> |
| Additional notes | <p>This lab reinforced the importance of understanding network interfaces, filtering traffic efficiently, and analyzing packet details for signs of anomalies. Using -nn to disable name resolution is a best practice to avoid alerting threat actors. Viewing packets in hex/ASCII helps analysts detect patterns, malware signatures, and suspicious payloads. Future steps include practicing more advanced TCPDump filters and correlating packet data with SIEM logs for deeper incident investigations.</p>                                                                                                                                                                                                                                                                                                                                                                                             |

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| <b>Date:</b><br>May 29, 2026. | <b>Entry:</b><br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>            | Investigated a suspicious file hash using VirusTotal to determine if it was malicious and identify indicators of compromise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Tool(s) used</b>           | <ul style="list-style-type: none"> <li>• VirusTotal (file hash analysis)</li> <li>• MITRE ATT&amp;CK framework (for TTP classification)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>The 5 W's</b>              | <p>Capture the 5 W's of an incident.</p> <ul style="list-style-type: none"> <li>• Who caused the incident: Malware family <i>FlagPro</i> is associated with APT activity.</li> <li>• What happened: A file hash (54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b) was flagged by 54/71 vendors as malicious.</li> <li>• When: Detected during the “Investigate a Suspicious File Hash” lab on May 29, 2026.</li> <li>• Where: VirusTotal online analysis platform.</li> <li>• Why: To identify IoCs and understand attacker techniques for persistence and evasion.</li> </ul>                                                                                                                                                                                            |
| <b>Additional notes</b>       | <p>This investigation reinforced the importance of using file hash analysis tools like VirusTotal to quickly identify malicious files and extract indicators of compromise (IoCs). The sample analyzed was confirmed to be a FlagPro backdoor trojan, detected by 54 out of 71 vendors. The behavior analysis revealed anti-debugging and lateral movement techniques, which align with MITRE ATT&amp;CK tactics such as T1497 (Virtualization/Sandbox Evasion). This exercise helped me practice identifying IoCs across multiple categories — hash values, tools, and TTPs — and understand how each fits into the Pyramid of Pain. It also emphasized how detection methods vary in difficulty to evade, and how combining multiple detection layers strengthens incident response.</p> |

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| <b>Date:</b><br>May 29, 2026 | <b>Entry:</b><br>5                                                                                                                                                                                                                                                                                                                                           |
| Description                  | Responded to a phishing alert (Ticket A-2703) using the phishing incident-response playbook. Investigated the email and confirmed a malicious attachment linked to the FlagPro backdoor malware.                                                                                                                                                             |
| Tool(s) used                 | <ul style="list-style-type: none"><li>• VirusTotal</li><li>• Phishing Incident Response Playbook</li></ul>                                                                                                                                                                                                                                                   |
| The 5 W's                    | <ul style="list-style-type: none"><li>• Who: Sender "Def Communications" &lt;76tgyuhh6tgfrt7tg.su&gt;</li><li>• What: Phishing email with malicious attachment (FlagPro backdoor)</li><li>• When: July 20, 2022 09:30 AM (alert timestamp) Where: Corporate mail server (SERVER-MAIL)</li><li>• Why: Attempted malware delivery via phishing email</li></ul> |
| Additional notes             | This alert was escalated per playbook guidance after VirusTotal confirmed the file hash was malicious (54/71 detections). The exercise reinforced how phishing alerts are validated and escalated efficiently. It also highlighted the importance of checking sender domains and attachment hashes before closing alerts.                                    |

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| <b>Date:</b><br>May 29, 2026 | <b>Entry:</b><br>6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Description</b>           | In this lab, I examined how Suricata detects malicious or suspicious network activity by analyzing a PCAP file using a custom Suricata rule. I reviewed how Suricata generates alerts, how rule components work, and how to interpret Suricata's two primary log files: fast.log and eve.json. I also practiced extracting and correlating event data using jq.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Tool(s) used</b>          | Suricata, custom.rules, sample.pcap, fast.log, eve.json, jq                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>The 5 W's</b>             | <ul style="list-style-type: none"> <li>• <b>Who:</b> Myself, acting as the analyst, using Suricata to inspect network traffic.</li> <li>• <b>What:</b> Loaded a custom Suricata rule designed to alert on HTTP GET requests. Ran Suricata against a provided sample.pcap file. Generated and analyzed Suricata alerts in fast.log and eve.json. Used jq to parse JSON logs and extract specific fields such as timestamp, flow_id, signature, protocol, and destination IP. Correlated events using flow_id.</li> <li>• <b>When:</b> May 29, 2026, during the Suricata detection and response lab.</li> <li>• <b>Where:</b> Linux VM environment provided in the Coursera/Google Skills lab.</li> <li>• <b>Why:</b> To understand how Suricata rules trigger alerts, how logs are structured, and how analysts use Suricata output for detection, investigation, and correlation.</li> </ul> |
| <b>Additional notes</b>      | This lab strengthened my understanding of IDS/IPS behavior, signature-based detection, and log analysis workflows. It also reinforced how important it is to correlate events using flow IDs and how Suricata fits into a modern detection pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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**Reflections:** This portfolio activity helped me connect theory to practice across multiple labs. I gained hands-on experience investigating incidents, analyzing logs, and using tools like Suricata and SIEM to detect threats. It strengthened my understanding of how network telemetry, signatures, and alerts work together to identify malicious activity. Overall, this journal improved my confidence in applying detection and response concepts to real-world cybersecurity scenarios.